

HIMADRI S. PANDEY

Ph.D. Candidate, Machine Learning

Georgia Institute of Technology

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Research Interests

I study how organizations should allocate scarce resources, such as clinician attention, diagnostic time, and monitoring capacity, when decisions are made repeatedly, under uncertainty, and with imperfect data. My work in **healthcare operations** and **data-driven decision-making** combines optimization and sequential decision models with machine learning trained for the decision it supports, not only for predictive accuracy. I build these methods with practitioners, including Children's Healthcare of Atlanta and the NCAA-DoD CARE Consortium.

Education

Georgia Institute of Technology

Expected 2027

Ph.D. in Machine Learning, School of Industrial and Systems Engineering (ISyE)

University of Cincinnati

2019

B.S. in Computer Science, Minor in Mathematics and Physics | University Honors Scholar

Selected Honors and Awards

- Finalist | Lee B. Lusted Student Prize Competition, SMDM Annual Meeting 2026
- Finalist | 3 Minute Thesis Competition, Georgia Tech 2026
- Phillip J. and Delores A. Scott Graduate Student Health and Wellness Award | Georgia Tech 2024, 2025, 2026
- George Fellowship | ISyE, Georgia Tech 2024, 2025
- Winner | MIF Student Poster Competition, INFORMS 2024
- David Cowan Scholarship | Georgia Healthcare Information and Management Systems Society 2024

Publications

Peer-Reviewed

- **Pandey, H.S.**, Garcia, G.G.P., McCrea, M.A., McAllister, T.W., Broglio, S.P., Pasquina, P.F., Lempke, L., & CARE Consortium Investigators. "So Many Tests, So Little Time: Operationally Constrained Machine Learning for Diagnostic Assessment under Time Constraints." *IIE Transactions*, 2026.
- **Pandey, H.S.**, Schmidt, J.D., Lynall, R.C., McCrea, M.A., McAllister, T.W., Broglio, S.P., Pasquina, P.F., Garcia, G.G.P., & CARE Consortium Investigators. "Quantifying the Diagnostic Utility of Baseline Testing: An Analysis of Collegiate Athletes from the NCAA-DoD CARE Consortium Dataset." *The American Journal of Sports Medicine*, 53(1):181–191, 2025.
- Lempke, L.B., Boltz, A.J., Garcia, G.G.P., Syrydiuk, R.A., **Pandey, H.S.**, Pasquina, P.F., McCrea, M.A., McAllister, T.W., Broglio, S.P., & CARE Consortium Investigators. "Optimizing Baseline and Post-Concussion Assessments Through Factor Structure Analysis: Findings from the NCAA-DoD CARE Consortium." *The Clinical Neuropsychologist*, 2023.

Under Review / In Preparation

- **Pandey, H.S.**, Mai, M.V., Muthu, N., Wang, K., & Garcia, G.G.P. "Who Gets Watched? Allocating Scarce Escalation Resources on Pediatric Wards with Neural Index Policies." *Under review at Manufacturing & Service Operations Management (M&SOM)*.
- **Pandey, H.S.**, Garcia, G.G.P., "Rank, Then Allocate: A Sequential Knapsack Framework for Feature-Driven Personalization." *In Preparation for submission to MSOM*.

Book Chapters

- Lee, S.J., **Pandey, H.S.**, & Garcia, G.G.P. "Designing Interpretable Machine Learning Models Using Mixed Integer Programming." *Encyclopedia of Optimization*, 3rd ed., Springer, 2023.

Conference Proceedings & Preprints

- Schreiner, H., **Pandey, H.S.**, Sokoloff, M., Tomko, K., Hasse, C., & Hittle, B. “A Python Upgrade to the GooFit Package for Parallel Fitting.” *CHEP 2018 Proceedings*, 2018.
- Schreiner, H., Hasse, C., Hittle, B., **Pandey, H.S.**, Sokoloff, M., & Tomko, K. “GooFit 2.0.” *arXiv preprint*, 2017.

Presentations

Refereed Talks

- **Pandey, H.S.**, Wang, K., & Garcia, G.G.P. “Resource-Aware Decision Making for Pediatric Post-Surgical Monitoring via Neural Index Policies.” *Society for Medical Decision Making (SMDM) Annual Meeting*, 2026.
- **Pandey, H.S.**, Schmidt, J.D., Lynall, R.C., McCrea, M.A., McAllister, T.W., Broglio, S.P., Pasquina, P.F., Garcia, G.G.P., & CARE Consortium Investigators. “Value of Baseline Testing in Concussion Diagnosis.” *American College of Sports Medicine (ACSM) Annual Meeting*, 2024.
- **Pandey, H.S.** “Time Constrained Concussion Diagnosis: An Analysis of the NCAA-DoD CARE Consortium Dataset” *Poster, MIF Student Poster Competition (Winner), INFORMS Annual Meeting*, Seattle, WA, 2024.
- Schreiner, H., **Pandey, H.S.**, Sokoloff, M., Tomko, K., Hasse, C., & Hittle, B. “A Python Upgrade to the GooFit Package for Parallel Fitting.” *CHEP 2018 (Computing in High-Energy Physics)*, Sofia, Bulgaria, 2018.

Invited Talks

- **Pandey, H.S.**, Wang, K., & Garcia, G.G.P. “Neural Index Policies for Multi-Action Restless Multi-Armed Bandits with Heterogeneous Budgets.” *INFORMS Healthcare Conference*, 2026.
- **Pandey, H.S.**, Garcia, G.G.P., Lempke, L., McCrea, M.A., McAllister, T.W., Broglio, S.P., Pasquina, P.F., & CARE Consortium Investigators. “So Many Tests, So Little Time: Operationally Constrained Machine Learning for Diagnostic Assessment under Time Constraints.” *INFORMS Healthcare Conference*, 2026.
- **Pandey, H.S.**, Wang, K., & Garcia, G.G.P. “Neural Index Policies for Restless Multi-Action Bandits with Heterogeneous Budgets.” *INFORMS Optimization Society Conference*, 2026.
- **Pandey, H.S.**, Wang, K., & Garcia, G.G.P. “Neural Index Policies for Restless Multi-Action Bandits with Heterogeneous Budgets.” *INFORMS Annual Meeting*, 2025.
- **Pandey, H.S.**, *INFORMS “Smarter Decisions for a Better World” Flash Research Talks*, Morehouse College, Atlanta, GA, Sept. 2025.
- **Pandey, H.S.** & Garcia, G.G.P. “Resource Allocation.” *INFORMS Annual Meeting*, 2024.
- **Pandey, H.S.**, Garcia, G.G.P., Lempke, L., McCrea, M.A., McAllister, T.W., Broglio, S.P., & CARE Consortium Investigators. “Optimizing Concussion Assessment: A Mixed-Integer Optimization Approach with Time Constraints.” *INFORMS Annual Meeting*, 2023.
- **Pandey, H.S.** “Using GooFit with Python.” *DIANA-HEP Workshop*, 2018.

Professional Experience

Applied Scientist Intern, *Amazon Web Services, Seattle, WA*

May 2025 – Aug. 2025

- Designed AWS’s first time-series forecasting framework for ML traffic between EC2 and S3 GPU fabrics, supporting long-term capacity planning.
- Cleaned and engineered large-scale, anomaly-prone datasets and incorporated multiple exogenous features into time-series models; presented results and insights directly to AWS customers.

Applied Scientist, *Lucas Systems, Pittsburgh, PA*

Aug. 2019 – June 2022

- Developed and deployed solutions for warehouse operations — a heuristics- and regression-based **online order-batching system**, a **real-time workforce forecasting tool**, and a heuristics-based **storage-location assignment strategy** — collectively reducing average picking time by 26%.
- Built real-time data processing and ML deployment pipelines on Azure and AWS.
- Mentored two data scientists and organized knowledge-sharing talks (“Machine Learning 101,” “Docker 101”).

DIANA-HEP Fellow, *CERN, Geneva, Switzerland*

April 2018 – June 2018

- Developed linear regression, Kalman filter, and machine learning models to implement track reconstruction.

AI Research Scholar, *MIT, Cambridge, MA*

Jan. 2018 – March 2018

- Researched into ways of replacing computationally expensive parts of track reconstruction with artificial intelligence in LHCb experiment at CERN.

Software Engineering Intern, <i>Pumpkin Inc, San Francisco, CA</i>	<i>Jan. 2017 – March 2017</i>
• Developed flight onboard software for a Cubesat. • Redesigned and built the company website.	

Leadership and Service

Conference Service	
Session Chair, <i>INFORMS Annual Meeting</i>	<i>2024, 2026</i>
Session Chair, <i>IISE Annual Meeting</i>	<i>2025</i>
Editorial and Reviewing	
Conference Abstract Reviewer, <i>SMDM Annual Meeting</i>	<i>2026</i>
Journal Reviewer, <i>BMJ Open Sport & Exercise Medicine</i>	<i>2025</i>
Outreach	
Invited Speaker — OR Outreach, <i>INFORMS Flash Research Talks, Morehouse College</i>	<i>Sept. 2025</i>
University Service	
Graduate Representative, <i>Emerging Leader Advisory Board, Georgia Tech</i>	<i>2024 – 2025</i>
Bee Well Ambassador, <i>ISyE, Georgia Tech</i>	<i>2023 – Present</i>
Member, <i>Mental Health Joint Allocation Committee, Georgia Tech</i>	<i>2023 – 2024</i>
Mental Health Committee Member, <i>Graduate Student Government Association, Georgia Tech</i>	<i>2023 – 2025</i>
Student Organization Leadership	
President, <i>UC CubeCats, University of Cincinnati</i>	<i>May 2017 – May 2018</i>

Teaching and Mentoring

Instructor – Machine Learning, <i>Georgia Institute of Technology</i>	<i>Summer 2026</i>
Peer Leader, <i>University of Cincinnati</i>	<i>Aug. 2016 – Dec. 2016</i>
Mentor, <i>CATiSE, UC CubeCats, University of Cincinnati</i>	<i>Aug. 2016 – Dec. 2016</i>
Coding Mentor, <i>Google Ignite CS Bearcat Coders, University of Cincinnati</i>	<i>Aug. 2015 – Dec. 2016</i>

Honors, Awards, and Fellowships

• Finalist <i>Lee B. Lusted Student Prize Competition, SMDM Annual Meeting</i>	<i>2026</i>
• Finalist <i>3 Minute Thesis Competition, Georgia Institute of Technology</i>	<i>2026</i>
• Phillip J. and Delores A. Scott Graduate Student Health and Wellness Award <i>Georgia Tech</i>	<i>2024, 2025, 2026</i>
• George Fellowship <i>ISyE, Georgia Institute of Technology</i>	<i>2024, 2025</i>
• Winner <i>MIF Student Poster Competition, INFORMS</i>	<i>2024</i>
• David Cowan Scholarship <i>Georgia Healthcare Information and Management Systems Society</i>	<i>2024</i>
• Topper Fellowship <i>Georgia Institute of Technology</i>	<i>2022</i>
• Employee of the Quarter (Q3) <i>Lucas Systems</i>	<i>2021</i>
• UC CEAS International Outreach Scholarship <i>University of Cincinnati</i>	<i>2015–2019</i>
• Best Senior Design Project (1st Prize) <i>CEAS-Expo, University of Cincinnati</i>	<i>2019</i>
• Dean’s List (7/8 semesters) <i>University of Cincinnati, CEAS</i>	<i>2015–2018</i>
• CRA-W Grace Hopper Celebration Scholar	<i>2018</i>
• DIANA-HEP Fellowship	<i>2018</i>
• SmallSat Student Award <i>Small Satellite Conference</i>	<i>2016, 2017</i>

Technical Skills

Programming: Python, C++, Julia, SQL, Java, MATLAB

ML / Optimization Frameworks: PyTorch, TensorFlow, scikit-learn, Gurobi, JuMP

Methods: Decision-focused learning, differentiable optimization, reinforcement learning, POMDPs, mixed-integer programming, probabilistic models, deep learning

Tools / Infrastructure: Git/GitHub/GitLab, Docker, AWS, Azure, CMake, L^AT_EX